Deep Generative Models

A theory guide for Homework 1 and Homework 2

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Lecture numbers refer to the decks in `lectures/`.
Slide numbers are the numbers printed on the slides.

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Chapter 1

How to use this guide

This note teaches the mathematics that Homeworks 1 and 2 actually use. Each chapter states a definition, gives the intuition, derives the identity, draws the picture, and then poses a short practice problem with a worked solution. The practice problems are smaller than the homework. They train the same moves.

Intuition: What to read for which homework

Homework 1 uses Chapters 2–7. Homework 2 uses those chapters and then Chapters 8–10. The conditional ELBO, the hierarchical ELBO, dequantization, and the Glow convolution are extensions of the lectures. Their derivations are written out here.

Homework piece	Read	Slides
Gaussians, products, conditionals	Ch. 5	L2, 12–35
KL, Jensen–Shannon, perplexity	Ch. 4	L2, 36–42
Monte Carlo, importance sampling	Ch. 6	L2, 34–35; L4, 23–29
Markov models and autoregression	Ch. 7	L3
ELBO, reparameterization, CVAE	Ch. 8	L4, 12–40, 56–84
Two-layer hierarchical ELBO	Ch. 9	pattern of L4, 38–39
Coupling, flow prior, dequantization, Glow	Ch. 10	L5, 3–73

Definition: Notation used throughout

Random variables are capitals when the distinction matters, and the same letter is used for a value when the slides do so. A Gaussian $\mathcal{N}(x \mid \mu, \sigma^2)$ has variance σ^2, matching Lecture 2. log is the natural logarithm, so information quantities are in nats. $\mathrm{KL}(q\|p)$ means we use q where the reference in the denominator is p. The Jacobian $J_f(u)$ is the matrix of partial derivatives $\partial f_i / \partial u_j$.

Chapter 2

Probability

2.1 Events, variables, and densities

Definition: Density and mass

A discrete random variable X has a probability mass function $p(x) \geq 0$ with $\sum_x p(x) = 1$. A continuous random variable on $\mathbb{R}^d$ has a density $p(x) \geq 0$ with $\int p(x)\,dx = 1$. The density is not a probability. $p(x)$ can exceed 1. The probability of a set A is $\int_A p(x)\,dx$.

Intuition

Think of a density as height and a probability as area. A very narrow bump can be very tall and still have area one. This is why, later, the flow density $p_\theta(x)$ at an integer pixel is not the probability of that pixel.

The joint density $p(x, z)$ describes the pair. The marginal of x sums or integrates the other variable out,

$$p(x) = \int p(x, z)\,dz.$$

The conditional is the joint divided by the marginal, when the marginal is positive:

$$p(z \mid x) = \frac{p(x, z)}{p(x)}, \qquad p(x, z) = p(x \mid z)\,p(z).$$

The second display is the chain rule. It is the only legal way to build a joint from conditionals on a directed graph: each node is conditioned on its parents, and the graph is acyclic.

Result: Bayes

$$p(z \mid x) = \frac{p(x \mid z)\,p(z)}{p(x)}.$$

The denominator is the marginal likelihood. In a latent-variable model it is the integral that the ELBO avoids computing.

2.2 Expectation, variance, and Jensen

Definition: Expectation

For a function f and a density p,

$$\mathbb{E}_{x\sim p}[f(x)] = \int f(x)\, p(x)\, dx,$$

when the integral exists. Linearity holds even if the coordinates of x are dependent: $\mathbb{E}[a^\top x] = a^\top \mathbb{E}[x]$.

Variance is $\mathrm{Var}(f) = \mathbb{E}[f^2] - (\mathbb{E}f)^2$. For a sum of independent random variables the variances add. They do not add when the variables share randomness. That fact is the whole of the importance-sampling variance story: a bad proposal makes $f\, p/q$ wildly dependent on the draw.

Result: Jensen

If ψ is concave and the expectations exist, then $\psi(\mathbb{E}[X]) \geq \mathbb{E}[\psi(X)]$. The logarithm is concave, so

$$\log \mathbb{E}[X] \geq \mathbb{E}[\log X].$$

Equality holds when X is almost surely constant, or more generally when ψ is affine on the range of X.

Intuition

Jensen is the reason a lower bound appears every time a logarithm moves inside an expectation. Lecture 4, slide 34, writes exactly this for $\psi = \log$. Homework 2 uses it twice: once for the ELBO, and once for variational dequantization.

Practice: A one-line Jensen gap

Let $Z \in \{0, 1\}$ with equal probability, and let $f(0) = 1$, $f(1) = e$. Compare $\log \mathbb{E}[f(Z)]$ with $\mathbb{E}[\log f(Z)]$.

Worked solution

$\mathbb{E}[f] = (1 + e)/2$, so $\log \mathbb{E}[f] = \log(1 + e) - \log 2$. And $\mathbb{E}[\log f] = \frac{1}{2}\log 1 + \frac{1}{2}\log e = \frac{1}{2}$. The gap is $\log(1 + e) - \log 2 - \frac{1}{2} > 0$ because f is not constant.

Chapter 3

Matrix calculus and Jacobians

Generative models differentiate scalars with respect to vectors and matrices. This chapter fixes the conventions and derives the four derivatives the homeworks need: a linear form, a quadratic form, a log-determinant, and a Jacobian determinant.

3.1 Layout

A scalar $L(\theta)$ has a gradient $\nabla_\theta L$ that is a column vector of the same shape as θ. The differential is the inner product

$$dL = \langle \nabla_\theta L,\, d\theta \rangle.$$

If θ is a matrix, the inner product is the Frobenius product $\langle A, B \rangle = \operatorname{tr}(A^\top B)$. This is the layout used below. It matches the chain rule in coordinates.

Result: Linear and quadratic forms

Let $a \in \mathbb{R}^d$ be constant and let $A \in \mathbb{R}^{d \times d}$ be constant. Then

$$\nabla_x (a^\top x) = a,$$
$$\nabla_x (x^\top A x) = (A + A^\top) x.$$

If A is symmetric, the second gradient is $2Ax$. In particular $\nabla_x \|x\|^2 = 2x$ and $\nabla_x \frac{1}{2}\|x\|^2 = x$.

Intuition

The factor 2 is the usual one from $(x^2)' = 2x$. People drop it when they differentiate $\frac{1}{2} x^\top \Sigma^{-1} x$, because the $\frac{1}{2}$ cancels the 2. Gaussian exponents are written with that $\frac{1}{2}$ for this reason.

3.2 Trace and log-determinant

Result: Derivatives in the matrix

For constant B and invertible A,

$$\nabla_A \operatorname{tr}(A^\top B) = B,$$
$$\nabla_A \log|\det A| = A^{-\top}.$$

The second identity is the matrix calculus behind Glow. If W depends on a diagonal vector s only through $W = \operatorname{diag}(s)$, then

$$\frac{\partial}{\partial s_i} \log|\det W| = \frac{1}{s_i} \quad \text{when } s_i \neq 0,$$

and the gradient of $\sum_i \log|s_i|$ is the vector $(1/s_i)$.

A short derivation of the log-determinant: $\det(A + dA) = \det(A)\det(I + A^{-1}dA)$, and $\det(I + H) = 1 + \operatorname{tr}(H) + o(\|H\|)$. Therefore

$$d \log|\det A| = \operatorname{tr}(A^{-1}\, dA) = \langle A^{-\top},\, dA \rangle.$$

3.3 Jacobians and the change-of-variables factor

Let $f : \mathbb{R}^D \to \mathbb{R}^D$ be continuously differentiable and invertible, with differentiable inverse, and with $J_f(u)$ invertible at the points we use. If u has density p_0 and $y = f(u)$, then y has density

$$p(y) = p_0(f^{-1}(y))\,|\det J_{f^{-1}}(y)| = p_0(u)\,|\det J_f(u)|^{-1}. \tag{3.1}$$

In logarithms, with $u = f^{-1}(y)$,

$$\log p(y) = \log p_0(u) - \log|\det J_f(u)|.$$

Intuition: Why the absolute value

A reflection $f(u) = -u$ preserves probability but has Jacobian -1. Densities are nonnegative, so the volume factor is the absolute value of the determinant. On a triangular matrix the determinant is the product of the diagonal entries, which is why coupling layers are easy.

Result: Triangular determinant

If L is lower triangular with ones on the diagonal, $\det L = 1$. If U is strictly upper triangular, then $U + \operatorname{diag}(s)$ is upper triangular with diagonal s, and $\det(U + \operatorname{diag}(s)) = \prod_i s_i$. A permutation

matrix P has $\det P = \pm 1$. Hence if $W = PL(U + \operatorname{diag}(s))$,

$$\log|\det W| = \sum_{i=1}^{C} \log|s_i|.$$

Practice: A 2×2 Jacobian

Let $f(u_1, u_2) = (u_1,\ u_2 + u_1^2)$. Compute J_f, its determinant, and the density of $y = f(u)$ when $u \sim \mathcal{N}(0, I_2)$.

Worked solution

$$J_f(u) = \begin{pmatrix} 1 & 0 \\ 2u_1 & 1 \end{pmatrix}, \qquad \det J_f = 1.$$

The map is a shear. It preserves volume, so $\log p(y) = \log p_0(f^{-1}(y))$. The inverse is $(y_1,\ y_2 - y_1^2)$, and

$$p(y) = (2\pi)^{-1} \exp\!\left(-\tfrac{1}{2} y_1^2 - \tfrac{1}{2}(y_2 - y_1^2)^2\right).$$

This is the map in Homework 2, problem 3.2. It will return there as a counterexample to a diagonal-Gaussian restriction.

Chapter 4

Divergences

4.1 Kullback–Leibler divergence

Definition

For densities p and q with $q > 0$ wherever $p > 0$,

$$\mathrm{KL}(p\|q) = \mathbb{E}_{x\sim p}\left[\log \frac{p(x)}{q(x)}\right] = \int p(x) \log \frac{p(x)}{q(x)}\, dx.$$

Result: Properties from Lecture 2, slides 36–39

1. $\mathrm{KL}(p\|q) \geq 0$.
2. $\mathrm{KL}(p\|q) = 0$ if and only if $p = q$ almost everywhere.
3. $\mathrm{KL}(p\|q) \neq \mathrm{KL}(q\|p)$ in general.
4. KL is not a distance: it fails symmetry and the triangle inequality.
5. KL may be infinite when the supports disagree.

The proof of nonnegativity is Jensen. Write

$$-\mathrm{KL}(p\|q) = \mathbb{E}_p\left[\log(q/p)\right] \leq \log \mathbb{E}_p[q/p] = \log \int q = 0,$$

so the KL is at least zero. Equality in Jensen holds when q/p is constant on the support of p. Normalization forces that constant to be 1.

In a VAE the bound contains $\mathrm{KL}(q_\phi(z \mid x) \| p_\theta(z))$. The first argument is the encoder, the second is the prior. Pushing that KL to zero makes the encoder match the prior, which is not the same as making the prior match the aggregated encoder. The asymmetry is the source of posterior collapse and of the off-mode mass of a Gaussian prior.

4.2 Closed-form Gaussian KL

Let $q = \mathcal{N}(\mu_q, \mathrm{diag}(\sigma_q^2))$ and $p = \mathcal{N}(\mu_p, \mathrm{diag}(\sigma_p^2))$ on $\mathbb{R}^d$, with every $\sigma > 0$. Independence across coordinates splits the KL into a sum. In one dimension, the Gaussian density contributes

$$\log q(z) = -\tfrac{1}{2}\log(2\pi\sigma_q^2) - \frac{(z-\mu_q)^2}{2\sigma_q^2}.$$

Taking the expectation under q and subtracting the same expression for p gives, after using $\mathbb{E}_q[(z-\mu_q)^2] = \sigma_q^2$ and $\mathbb{E}_q[(z-\mu_p)^2] = \sigma_q^2 + (\mu_q - \mu_p)^2$,

Result: Diagonal Gaussian KL

$$\mathrm{KL}(q\|p) = \frac{1}{2}\sum_{j=1}^{d}\left(\log\frac{\sigma_{p,j}^2}{\sigma_{q,j}^2} - 1 + \frac{\sigma_{q,j}^2 + (\mu_{q,j}-\mu_{p,j})^2}{\sigma_{p,j}^2}\right). \tag{4.1}$$

Against the standard normal $p = \mathcal{N}(0, I)$ this collapses to

$$\mathrm{KL}(q\|\mathcal{N}(0,I)) = \frac{1}{2}\sum_{j=1}^{d}\left(\mu_{q,j}^2 + \sigma_{q,j}^2 - 1 - \log\sigma_{q,j}^2\right). \tag{4.2}$$

Equation (4.2) is the KL implemented in the VAE notebook. Equation (4.1) is Homework 2, problem 1.3, before that specialization.

Practice: A number you can check by hand

Compute $\mathrm{KL}(\mathcal{N}(0,1)\|\mathcal{N}(0,e))$ and $\mathrm{KL}(\mathcal{N}(0,e)\|\mathcal{N}(0,1))$.

Worked solution

Use $d = 1$, $\mu = 0$. First pair: $\sigma_q^2 = 1$, $\sigma_p^2 = e$, so $\frac{1}{2}(\log e - 1 + 1/e) = \frac{1}{2}(1 - 1 + e^{-1}) = 1/(2e)$. Second pair: $\sigma_q^2 = e$, $\sigma_p^2 = 1$, so $\frac{1}{2}(\log e^{-1} - 1 + e) = \frac{1}{2}(-1 - 1 + e) = (e-2)/2$. They differ. KL is not symmetric, and the direction that puts a wide Gaussian on a narrow one is the more expensive one here.

4.3 Jensen–Shannon divergence

Let $m = \frac{1}{2}(p + q)$. Lecture 2 defines

$$\mathrm{JS}(p\|q) = \tfrac{1}{2}\mathrm{KL}(p\|m) + \tfrac{1}{2}\mathrm{KL}(q\|m).$$

JS is symmetric. In nats, $0 \le \mathrm{JS}(p\|q) \le \log 2$. The upper bound 1 on the lecture slides is the same statement in bits, where the log is base 2. In nats the sharp cap is $\ln 2$, not 1. $\sqrt{\mathrm{JS}}$ is a metric. JS itself is not, because it can violate the triangle inequality.

Intuition

KL is infinite when supports are disjoint, so it cannot compare a model that puts no mass on a point with a data distribution that does. JS stays finite because m charges both supports. GANs later exploit a divergence with that behaviour. Homework 1 asks you to see the failure of KL directly.

Chapter 5

The Gaussian family

5.1 Affine images

Result: Homework 1, the basic transform

If $x \sim \mathcal{N}(\mu, \Sigma)$ and $y = Ax + b$, then

$$y \sim \mathcal{N}(A\mu + b,\ A\Sigma A^\top),$$

on the affine image of the column space. If $A\Sigma A^\top$ is singular, y has no density on the ambient Euclidean space. It has a density on the affine hull of its support.

The moment proof is two lines. $\mathbb{E}[y] = A\mu + b$ by linearity. The covariance is

$$\mathbb{E}[(y - \mathbb{E}y)(y - \mathbb{E}y)^\top] = A\,\mathbb{E}[(x - \mu)(x - \mu)^\top]\,A^\top = A\Sigma A^\top.$$

A Gaussian is determined by its mean and covariance, so the law is that Gaussian. The same calculation with A a row vector is the reparameterization trick in one coordinate: if $\varepsilon \sim \mathcal{N}(0, 1)$, then $\mu + \sigma\varepsilon \sim \mathcal{N}(\mu, \sigma^2)$.

Practice

Let $x \sim \mathcal{N}(1, 4)$ and $y = 3x - 2$. Find $\mathbb{E}[y]$ and $\mathrm{Var}(y)$.

Worked solution

Mean $3 \cdot 1 - 2 = 1$. Variance $3^2 \cdot 4 = 36$. So $y \sim \mathcal{N}(1, 36)$.

5.2 Conditionals and products

For a joint Gaussian, the conditional of one block given another is Gaussian. Its mean is linear in the observation, and its covariance does not depend on the observation. That is why a linear-Gaussian encoder can be written in closed form, and why a neural encoder is the nonlinear replacement of that conditional mean and variance.

The product of two Gaussian densities, as a function of x, is an unnormalized Gaussian whose precision (inverse covariance) is the sum of the precisions. The normalizing constant of that product is itself a Gaussian density in the difference of the means. Homework 1 asks this identity because it is Bayes' rule for a Gaussian likelihood and a Gaussian prior: posterior precision equals prior precision plus likelihood precision.

5.3 Mixtures

A mixture $p(x) = \sum_{k=1}^{K} \pi_k \, \mathcal{N}(x \mid \mu_k, \Sigma_k)$ with $\pi_k > 0$ and $\sum_k \pi_k = 1$ is nonnegative and integrates to one. Its mean is $\sum_k \pi_k \mu_k$. Its covariance is not the mixture of the covariances alone. The law of total variance adds the covariance of the component means:

$$\operatorname{Cov}(x) = \mathbb{E}[\operatorname{Cov}(x \mid k)] + \operatorname{Cov}(\mathbb{E}[x \mid k]).$$

A single Gaussian cannot represent well-separated modes. A mixture can. Lecture 4, slides 3–11, uses the iris petals as the picture. The VAE notebook, part II, trains that comparison in two dimensions.

5.4 Maximum entropy

Among densities on $\mathbb{R}^d$ with a fixed mean and a fixed covariance, the Gaussian has the largest entropy

$$h(p) = -\int p \log p.$$

For $\mathcal{N}(\mu, \Sigma)$ the entropy is $\frac{d}{2} \log(2\pi e) + \frac{1}{2} \log\det \Sigma$. Subtracting two such entropies, and accounting for the cross term that appears when the means differ, is one derivation of the multivariate KL. The diagonal formula (4.1) is the coordinate form of that identity.

Chapter 6

Monte Carlo

6.1 The sample average

Definition

If $x^{(1)}, \ldots, x^{(N)}$ are drawn independently from p, the Monte Carlo estimator of $\mathbb{E}_p[f]$ is

$$\hat{f}_N = \frac{1}{N} \sum_{n=1}^{N} f(x^{(n)}).$$

It is unbiased. If $\mathrm{Var}_p(f) = \sigma_f^2 < \infty$, then $\mathrm{Var}(\hat{f}_N) = \sigma_f^2 / N$.

Intuition

Unbiased means the average of the estimator, over datasets of samples, equals the integral. It does not mean one sample is exact. The error typically shrinks like $1/\sqrt{N}$, because the standard deviation is $\sigma_f / \sqrt{N}$. Lecture 2, slides 34–35, and Homework 1, problem 2, turn that variance into a sample size through Chebyshev or a Chernoff bound. Chebyshev uses only the variance. A Chernoff bound uses the moment generating function and can demand fewer samples while remaining a guarantee.

6.2 Importance sampling

When draws from p are unavailable, draw $x \sim q$ instead and correct the weight:

$$\mathbb{E}_p[f] = \mathbb{E}_q\left[f(x)\frac{p(x)}{q(x)}\right],$$

provided q is positive wherever pf is nonzero. The weight p/q is the importance weight. The estimator is unbiased for any such q. Its variance is small when q is large where $|f|p$ is large, and it

is huge when q is tiny on that set.

Lecture 4, slides 23–29, shows a proposal that misses the posterior mass. The likelihood estimator then has rare enormous weights. The ELBO replaces that high-variance estimator by a bound whose gap is exactly a KL. You give up unbiasedness of the likelihood and gain a gradient you can train.

6.3 Perplexity

For a discrete model, perplexity is $\exp\big(-\mathbb{E}_{x\sim p_{\text{data}}} \log p_\theta(x)\big)$, or the same expression with the average over a finite test set. It is an exponential of a cross-entropy. Jensen applied to the exponential, or a bias expansion of the logarithm of an average, shows that a naive plug-in can be biased. Homework 1 asks for that expansion. The point to remember is that the log of an average is not the average of the logs.

Chapter 7

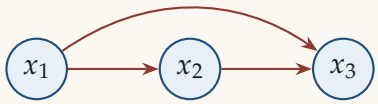
Autoregressive models

7.1 The chain rule on a sequence

Any positive joint mass on binary strings $x \in \{0,1\}^D$ factors as

$$p(x) = \prod_{d=1}^{D} p(x_d \mid x_{<d}),$$

with $x_{<1}$ empty. A fully general conditional at step d is a table of size 2^{d-1}. Summing the sizes gives $2^D - 1$ free parameters, the whole probability simplex. That model can represent anything and cannot be estimated from data once D is large.



An autoregressive factorization. Each variable is conditioned only on the variables already generated.

7.2 FVSBN and NADE

A fully visible sigmoid belief network replaces the table by a logistic regression on the past:

$$p(x_d = 1 \mid x_{<d}) = \sigma\Big(b_d + \sum_{k<d} W_{dk} x_k\Big).$$

Step d costs d parameters, and the total is $D(D+1)/2$. The map from parameters to distributions is injective for this linear-logit family, so that count is the dimension of the model.

NADE shares one hidden state across the conditionals,

$$h_d = \phi\Big(c + \sum_{k<d} W_{:,k} x_k\Big), \qquad p(x_d = 1 \mid x_{<d}) = \sigma(v_d^\top h_d + b_d).$$

Sharing W makes the parameter count linear in D rather than quadratic. The recurrence $a_{d+1} = a_d + W_{:,d} x_d$ evaluates every conditional without recomputing the sum from scratch. Real NADE uses the same hidden state and emits a Gaussian conditional, with mean linear in h_d and variance $\exp(s_d)$.

Intuition

Likelihood of a fully observed x can be computed in parallel across coordinates, because every past bit is already known. Sampling cannot. Coordinate d is drawn only after coordinates $1, \ldots, d-1$ exist. Lecture 3 says this in one line, and MAF inherits it.

7.3 MADE masks

MADE imposes the autoregressive constraint with binary masks so that one forward pass evaluates every conditional. Number the hidden units by the largest input index they may depend on. A hidden unit with index m may read inputs of index at most m. An output unit for coordinate d may read hidden units of index strictly less than d. The strict inequality at the output stops x_d from depending on itself.

Practice: A mask you can draw

For $D = 3$ and one hidden layer with indices $(1, 2, 2)$, say which inputs each output may use.

Worked solution

Output 1 may use hidden units of index < 1. There are none. So $p(x_1)$ is unconditional. Output 2 may use the hidden unit of index 1, which itself may see only x_1. Output 3 may use all three hidden units, hence x_1 and x_2, and not x_3. That is an autoregressive family.

7.4 Graphs

A directed acyclic graph defines a legal joint by multiplying one conditional per node, given its parents. A topological order is an order in which every parent appears before its child. Many orders can be legal. A directed cycle does not define a joint by the same multiplication: the conditionals refer to each other and need not be normalizable. A rooted forest is the special case in which each node has at most one parent. Markov chains are the path case, homogeneous when the conditional does not depend on time, inhomogeneous when it does.

The KL projection of a full joint onto an autoregressive family is the set of conditional tables of the true joint, in that order. The KL gap equals a conditional mutual information: the information about x_d that the discarded non-parents still carry. Maximum likelihood on a fully observed table estimates those conditionals by counts.

Chapter 8

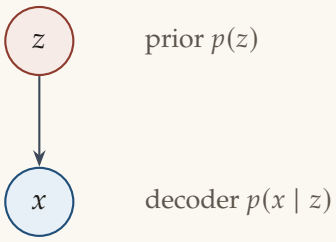
Variational autoencoders

8.1 The model

A latent-variable model says that each observation x was produced by a hidden z:

$$p_\theta(x, z) = p_\theta(x \mid z)\, p_\theta(z), \qquad p_\theta(x) = \int p_\theta(x \mid z)\, p_\theta(z)\, dz.$$

Generation draws z from the prior and then x from the decoder. The integral is the obstacle. It is the marginal likelihood.



For a conditional model the sheet adds a variable u, for example a digit label:

$$p_\theta(x, z \mid u) = p_\theta(x \mid z, u)\, p_\theta(z \mid u).$$

Every derivation below is the unconditional one with u written next to x. The pictures gain arrows from u into the prior and into the decoder.

8.2 Two derivations of the ELBO

Fix a density $q(z)$ that does not depend on θ, positive wherever the posterior is. The marginal is an expectation under q:

$$p_\theta(x) = \mathbb{E}_{z \sim q}\left[\frac{p_\theta(x, z)}{q(z)}\right].$$

Apply log and Jensen. The logarithm is concave, so

$$\log p_\theta(x) \geq \mathbb{E}_{z\sim q}\left[\log \frac{p_\theta(x,z)}{q(z)}\right] =: \mathcal{L}(x;\theta,q). \tag{8.1}$$

The right-hand side is the evidence lower bound.

The second derivation starts from the KL between q and the true posterior $p_\theta(z \mid x) = p_\theta(x,z)/p_\theta(x)$:

$$\begin{aligned}\mathrm{KL}(q(z)\|p_\theta(z \mid x)) &= \mathbb{E}_q\left[\log q(z) - \log p_\theta(x,z) + \log p_\theta(x)\right] \\ &= \log p_\theta(x) - \mathbb{E}_q\left[\log \frac{p_\theta(x,z)}{q(z)}\right].\end{aligned}$$

Rearrangement is the identity on Lecture 4, slides 38–39,

$$\log p_\theta(x) = \mathcal{L}(x;\theta,q) + \mathrm{KL}\big(q(z)\|p_\theta(z \mid x)\big). \tag{8.2}$$

Nonnegativity of KL gives $\mathcal{L} \leq \log p_\theta(x)$. The gap is zero if and only if $q(z) = p_\theta(z \mid x)$ almost everywhere.

Intuition

Jensen moves the log inside and produces a bound. The KL identity explains the gap. When you write Homework 2, problem 1.1, do both, and insert u into $p(x,z \mid u)$ and $q(z \mid x,u)$. Equality is the sentence "KL is zero only when the densities agree."

8.3 Reconstruction minus KL

Expand $p(x,z) = p(x \mid z)p(z)$ inside the ELBO:

$$\begin{aligned}\mathcal{L} &= \mathbb{E}_q[\log p_\theta(x \mid z) + \log p_\theta(z) - \log q(z)] \\ &= \mathbb{E}_q[\log p_\theta(x \mid z)] - \mathrm{KL}\big(q(z)\|p_\theta(z)\big).\end{aligned}$$

The first term rewards a decoder that regenerates x from the latent the encoder proposed. The second term penalizes an encoder that wanders away from the prior. For the conditional model,

$$\mathcal{L}(\theta,\phi;x,u) = \mathbb{E}_{q_\phi(z|x,u)}[\log p_\theta(x \mid z,u)] - \mathrm{KL}\big(q_\phi(z \mid x,u)\|p_\theta(z \mid u)\big). \tag{8.3}$$

Suppose $q(z \mid x,u) = p(z \mid u)$. The KL term in (8.3) is zero. The gap that remains is

$$\log p(x \mid u) - \mathcal{L} = \log \mathbb{E}_{p(z|u)}[p(x \mid z,u)] - \mathbb{E}_{p(z|u)}[\log p(x \mid z,u)].$$

That is Jensen's gap for the random variable $p(x \mid z, u)$ as z varies. It is strictly positive unless $p(x \mid z, u)$ does not depend on z.
If the decoder ignores z, so that $p(x \mid z, u) = p(x \mid u)$, Bayes' rule gives $p(z \mid x, u) = p(z \mid u)$. Matching the encoder to the prior then makes both the KL and the gap vanish. The latent carries no information about x. That is posterior collapse. The bound is tight and useless.

8.4 The reparameterization trick

The KL in (4.2) has a gradient in closed form. The reconstruction term does not. A score-function estimator differentiates through $\log q$ and has high variance. The reparameterization writes the sample as a deterministic function of the parameters and of noise that does not depend on them:

$$\varepsilon \sim \mathcal{N}(0, I), \qquad z = \mu_\phi(x) + \sigma_\phi(x) \odot \varepsilon,$$

where $\sigma = \exp(\frac{1}{2} \log \sigma^2)$ if the network emits a log-variance. Because ε is standard normal, z is exactly $\mathcal{N}(\mu, \operatorname{diag}(\sigma^2))$.

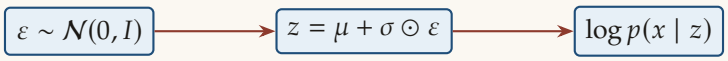
A Monte Carlo reconstruction objective with one sample is

$$\widehat{\mathcal{L}}_{\text{rec}}(\phi) = \log p_\theta\big(x \mid \mu_\phi + \sigma_\phi \odot \varepsilon\big).$$

The chain rule is

$$\nabla_\phi \widehat{\mathcal{L}}_{\text{rec}} = \frac{\partial \log p_\theta}{\partial z} \Big(\frac{\partial \mu}{\partial \phi} + \varepsilon \odot \frac{\partial \sigma}{\partial \phi} \Big).$$

Differentiate through the sample, not through a density score. Lecture 4, slides 60–64, is this argument.



parameters enter only here

For images in the notebook the decoder outputs Bernoulli logits, one per pixel. The reconstruction term is a binary cross-entropy with logits, summed over pixels and averaged over the batch. Do not apply a sigmoid before that loss. The sigmoid sits inside the stable implementation of the cross-entropy. With $\beta = 1$ the training objective per image is that cross-entropy plus the KL in (4.2).

Practice: Check the transform

Let $\varepsilon \sim \mathcal{N}(0, 1)$, $\mu = 2$, $\sigma = 3$. Find $\mathbb{E}[z]$ and $\operatorname{Var}(z)$ for $z = \mu + \sigma\varepsilon$. Then compute $\partial z / \partial \mu$ and $\partial z / \partial \sigma$.

Worked solution

$\mathbb{E}[z] = 2$ and $\mathrm{Var}(z) = 9$, so $z \sim \mathcal{N}(2, 9)$. The partial derivatives are $\partial z/\partial \mu = 1$ and $\partial z/\partial \sigma = \varepsilon$. A loss $L(z)$ therefore has $\partial L/\partial \sigma = L'(z)\,\varepsilon$.

Chapter 9

Hierarchical latents

One latent layer can be too weak a prior. A two-layer model draws a coarse code and then a fine code:

$$p_\theta(x, z_1, z_2) = p_\theta(z_2)\, p_\theta(z_1 \mid z_2)\, p_\theta(x \mid z_1),$$

with the conditional independence $x \perp z_2 \mid z_1$. The variational family used in the homework respects the same order,

$$q_\phi(z_1, z_2 \mid x) = q_\phi(z_2 \mid x)\, q_\phi(z_1 \mid z_2, x).$$



9.1 The hierarchical ELBO

Apply (8.2) to the pair (z_1, z_2). The gap is

$$\log p_\theta(x) - \mathcal{L}_H(x) = \mathrm{KL}\big(q(z_1, z_2 \mid x) \| p(z_1, z_2 \mid x)\big).$$

Expand the joint and the variational factorizations. The cross terms separate into three pieces:

$$\begin{aligned}\mathcal{L}_H(x) = \mathbb{E}_{q(z_1, z_2 \mid x)}[\log p(x \mid z_1)] - \mathrm{KL}\big(q(z_2 \mid x) \| p(z_2)\big) \\ - \mathbb{E}_{q(z_2 \mid x)}\Big[\mathrm{KL}\big(q(z_1 \mid z_2, x) \| p(z_1 \mid z_2)\big)\Big].\end{aligned}$$

The inner KL is averaged under $q(z_2 \mid x)$ because z_2 is still random when that conditional KL is evaluated. You draw the coarse code from the approximate posterior, then score the fine code.

9.2 Comparison with a one-layer bound

Let $q(z_1 \mid x)$ be the marginal of the variational joint, and define the ordinary ELBO that uses only z_1,

$$\mathcal{L}_M(x) = \mathbb{E}_{q(z_1|x)}[\log p(x \mid z_1)] - \mathrm{KL}\big(q(z_1 \mid x) \| p(z_1)\big).$$

A second KL decomposition, this time of $q(z_2 \mid z_1, x)$ against $p(z_2 \mid z_1)$, yields

$$\mathcal{L}_H(x) = \mathcal{L}_M(x) - \mathbb{E}_{q(z_1|x)}\Big[\mathrm{KL}\big(q(z_2 \mid z_1, x) \| p(z_2 \mid z_1)\big)\Big].$$

The subtracted term is nonnegative, so $\mathcal{L}_H \le \mathcal{L}_M$. Equality holds if and only if $q(z_2 \mid z_1, x) = p(z_2 \mid z_1)$ almost everywhere. For a fixed marginal $q(z_1 \mid x)$, that conditional is exactly the one that maximizes $\mathcal{L}_H$.

The marginal prior $p(z_1) = \int p(z_1 \mid z_2)p(z_2)\,dz_2$ may have no closed form. For an arbitrary positive auxiliary $r(z_2 \mid z_1)$,

$$\begin{aligned}\log p(z_1) = \mathbb{E}_r[\log p(z_1 \mid z_2)] - \mathrm{KL}(r(z_2 \mid z_1) \| p(z_2)) \\ + \mathrm{KL}(r(z_2 \mid z_1) \| p(z_2 \mid z_1)).\end{aligned}$$

Drop the last KL, which is nonnegative, and you have a lower bound. The bound is an equality when $r = p(z_2 \mid z_1)$. Therefore

$$\log p(z_1) = \sup_r \Big\{\mathbb{E}_r[\log p(z_1 \mid z_2)] - \mathrm{KL}(r \| p(z_2))\Big\},$$

and the supremum is attained at the true conditional. This is the same "gap equals a KL" pattern as (8.2).

Practice

In the decomposition of $\mathcal{L}_H$, why is the fine-code KL inside an expectation over $q(z_2 \mid x)$, rather than evaluated at a single z_2?

Worked solution

The KL $\mathrm{KL}(q(z_1 \mid z_2, x) \| p(z_1 \mid z_2))$ is a function of z_2. The variational joint draws z_2 from $q(z_2 \mid x)$ before drawing z_1. Taking the expectation is what the joint KL produces when you integrate that coordinate. Freezing one z_2 would describe a different, degenerate inference distribution.

Chapter 10

Normalizing flows

10.1 Change of variables as a model

Choose a simple base $u \sim p_0 = \mathcal{N}(0, I)$ and a stack of invertible maps $y = f(u)$. Equation (3.1) gives the exact density of y. There is no ELBO. Training maximizes the log-density of the data, which is a base log-density plus a sum of log-absolute-determinants, one per layer. Lecture 5, slides 3–34, builds this from equivalent events up to the chain rule for Jacobians:

$$\log|\det J_{f_K \circ \cdots \circ f_1}| = \sum_{k=1}^{K} \log|\det J_{f_k}|.$$

Intuition

A flow spends its capacity on maps whose Jacobian determinant is cheap. A generic neural network is invertible only by accident, and its determinant costs $O(D^3)$. Coupling layers, autoregressive maps, and 1×1 convolutions are the designs that keep the determinant in closed form.

10.2 Affine coupling

Split $h = (h_A, h_B)$ with dimensions $d_A + d_B = D$. RealNVP, Lecture 5 slides 52–64, sets

$$\begin{aligned} y_A &= h_A, \\ y_B &= h_B \odot \exp\big(s(h_A)\big) + t(h_A). \end{aligned}$$

The functions s and t are ordinary neural networks. They need not be invertible.

Result: Homework 2, problem 4.1

The map is invertible:

$$h_A = y_A,$$
$$h_B = \big(y_B - t(y_A)\big) \odot \exp\big(-s(y_A)\big).$$

Order the coordinates as (A, B). The Jacobian is block triangular,

$$J = \begin{pmatrix} I & 0 \\ * & \operatorname{diag}(e^{s(h_A)}) \end{pmatrix},$$

so $\log|\det J| = \sum_{j=1}^{d_B} s_j(h_A)$. Therefore

$$\log p(y) = \log p_0(f^{-1}(y)) - \sum_{j=1}^{d_B} s_j(y_A).$$

The networks s and t do not appear in the determinant. Only the scale values do. That is why a large network is allowed.

NICE is the special case $s \equiv 0$, an additive coupling. Then $\det J = 1$. A separate diagonal rescaling layer carries the change of volume. Alternating which half is updated, and inserting a permutation of channels between couplings, lets every coordinate be transformed.

Practice: Invert a 2-D coupling by hand

Let $s(h_1) = \log 2$ and $t(h_1) = h_1$. Forward-map $(h_1, h_2) = (1, 3)$. Then invert the result.

Worked solution

$y_1 = 1$ and $y_2 = 3 \cdot e^{\log 2} + 1 = 7$. The inverse reads $h_1 = y_1 = 1$ and $h_2 = (7 - 1) \cdot e^{-\log 2} = 3$. The log-absolute-determinant of the forward map is $\log 2$.

10.3 A flow as a VAE prior

Keep a decoder $p_\theta(x \mid z)$ and replace the prior by a flow. Draw $u \sim \mathcal{N}(0, I)$, set $z = f_\psi(u)$, and let p_ψ be the induced density of z. The ELBO is still reconstruction minus a KL,

$$\mathcal{L}_{\theta,\psi}(x) = \mathbb{E}_{q(z|x)}[\log p_\theta(x \mid z)] - \mathrm{KL}\big(q(z \mid x) \| p_\psi(z)\big),$$

and the change of variables writes the prior log-density as

$$\log p_\psi(z) = \log p_0(f^{-1}(z)) - \log|\det J_f(f^{-1}(z))|.$$

Equivalently, define a decoder on the base space by $\tilde{p}(x \mid u) = p(x \mid f(u))$. Integrating u against p_0 gives the same marginal $p(x)$ as integrating z against p_ψ. If $z \sim q(z \mid x)$ and $u = f^{-1}(z)$, the induced density is

$$r(u \mid x) = q(f(u) \mid x)\,|\det J_f(u)|,$$

and $\mathrm{KL}(q\|p_\psi) = \mathrm{KL}(r\|p_0)$. The two ELBOs agree when the variational densities are exact images of each other.

Restrict both variational families to diagonal Gaussians. Take $f(u_1, u_2) = (u_1, u_2 + u_1^2)$ and $q(z) = \mathcal{N}(0, I_2)$. Section 3 showed that $\det J_f = 1$ and that the induced density of $u = f^{-1}(z)$ is

$$r(u) \propto \exp\!\left(-\tfrac{1}{2}u_1^2 - \tfrac{1}{2}(u_2 + u_1^2)^2\right),$$

up to the constant of the standard normal. This is not a diagonal Gaussian: the coordinates are dependent. No diagonal-Gaussian encoder in u-space represents r. Optimizing over diagonal Gaussians in z and in u are different problems, even though the generative models are the same.

10.4 Masked autoregressive flows

An autoregressive map with $y_d = \mu_d(y_{<d}) + \sigma_d(y_{<d})\,\varepsilon_d$ is triangular. Its log-determinant is $\sum_d \log \sigma_d$. MADE masks evaluate every μ_d and σ_d in one pass. Likelihood is parallel. Sampling remains sequential, because y_d needs $y_{<d}$. That is Lecture 5, slides 65–73, and it is part III of the flow notebook: a two-dimensional MAF on Lab chroma. An inverse autoregressive flow swaps the cost. Sampling becomes parallel and likelihood becomes sequential. The notebook asks for a MAF, not an IAF.

10.5 From a density to a pixel probability

A flow defines a density on $\mathbb{R}^D$. An image $x \in \mathbb{Z}^D$ is discrete. Assign to x the bin $B_x = x + [0, 1)^D$. These bins partition $\mathbb{R}^D$ up to boundaries of measure zero. If y has density p_θ and $x = \lfloor y \rfloor$, then

$$P_\theta(x) = \int_{[0,1)^D} p_\theta(x + u)\,du.$$

Each value is nonnegative, and the sum over $x \in \mathbb{Z}^D$ equals $\int_{\mathbb{R}^D} p_\theta = 1$. So P_θ is a probability mass function. The number $p_\theta(x)$ is a density evaluated at the corner of the bin. It is not $P_\theta(x)$.

The integral has no closed form for a neural flow. Choose a dequantizer $q(u \mid x)$ supported on the

unit bin, strictly positive inside it. Importance sampling writes

$$P_\theta(x) = \mathbb{E}_{q(u|x)}\left[\frac{p_\theta(x+u)}{q(u \mid x)}\right].$$

Jensen produces the variational dequantization bound

$$\log P_\theta(x) \geq \mathbb{E}_q\left[\log \frac{p_\theta(x+u)}{q(u \mid x)}\right] = \mathbb{E}_q[\log p_\theta(x+u)] + H\big(q(\cdot \mid x)\big). \tag{10.1}$$

The first term wants $x + u$ where the flow density is high. The entropy stops q from collapsing to a point, which would throw away the integral.

Define the normalized restriction of the model to the bin,

$$r_\theta(u \mid x) = \frac{p_\theta(x+u)}{P_\theta(x)}, \qquad u \in [0,1)^D.$$

The same KL algebra as (8.2) gives

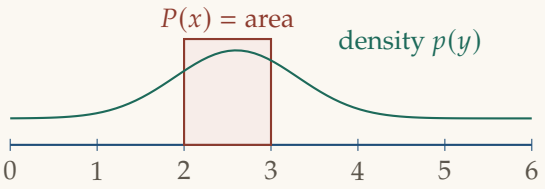
$$\log P_\theta(x) - \mathcal{L}_{\mathrm{DQ}} = \mathrm{KL}\big(q(u \mid x) \| r_\theta(u \mid x)\big).$$

The bound is tight if and only if $q = r_\theta(\cdot \mid x)$.

Uniform dequantization sets $q = 1$ on the bin. The bound becomes $\int_{[0,1)^D} \log p_\theta(x+u)\, du$, and it is tight if and only if $p_\theta(x+u)$ is constant on the bin. A conditional flow $u = g_\phi(\varepsilon; x)$, with $\varepsilon \sim p_0$ and $g_\phi(\cdot; x)$ a bijection onto $(0,1)^D$, substitutes the change-of-variables formula into (10.1):

$$\mathcal{L}_{\mathrm{DQ}} = \mathbb{E}_\varepsilon\Big[\log p_\theta\big(x + g_\phi(\varepsilon; x)\big) + \log|\det J_g(\varepsilon; x)| - \log p_0(\varepsilon)\Big].$$

A learned g can match a non-uniform r, so the KL gap can be smaller than the uniform gap.



The probability of an integer is the area of the density over the unit bin, not the height of the density.

10.6 Invertible 1×1 convolutions

RealNVP mixes channels with a fixed permutation. Glow learns the mix. For a tensor $X \in \mathbb{R}^{C \times H \times W}$,

$$Y_{:,i,j} = W X_{:,i,j}, \qquad W \in \mathbb{R}^{C \times C},$$

with the same W at every spatial site.

Result: Homework 2, problem 5

The map is invertible if and only if $\det W \neq 0$, and the inverse applies W^{-1} at each site. Vectorize the tensor in an order that keeps the C channels of each pixel together. The full Jacobian is block diagonal with HW blocks, each equal to W. Hence

$$\log|\det J| = HW \log|\det W|.$$

With the parameterization $W = PL\big(U + \operatorname{diag}(s)\big)$ of the previous determinant lemma,

$$\log|\det J| = HW \sum_{i=1}^{C} \log|s_i|.$$

The cost of the log-determinant is linear in the number of channels, not cubic, and it does not grow with a matrix factorization at every pixel.

Practice: One pixel, two channels

Let $H = W = 1$ and $W = \operatorname{diag}(2, 1/2)$. Compute $\log|\det J|$ two ways: from $HW \log|\det W|$, and from the sum of $\log|s_i|$.

Worked solution

$\det W = 1$, so $\log|\det J| = 0$. And $\log|2| + \log|1/2| = \log 2 + \log(1/2) = 0$. The volume of the two channels is unchanged: one axis stretches by 2, the other shrinks by 2.

Chapter 11

A short practice set

These five problems chain the moves above. Write the solution before you open the box.

Practice: Set, problem A

Let $q = \mathcal{N}(0, \mathrm{diag}(1, e))$ and $p = \mathcal{N}(0, I_2)$. Compute $\mathrm{KL}(q \| p)$ from (4.2).

Worked solution

Coordinate 1 contributes 0. Coordinate 2 has $\mu = 0$ and $\sigma^2 = e$, so $\frac{1}{2}(e - 1 - \log e) = \frac{1}{2}(e - 2)$. The sum is $(e - 2)/2$.

Practice: Set, problem B

Show that if $p(x \mid z) = p(x)$ does not depend on z, and $q(z) = p(z)$, then the ELBO equals $\log p(x)$.

Worked solution

The reconstruction term is $\mathbb{E}_q[\log p(x)] = \log p(x)$. The KL against the prior is zero. This is the collapsed case: the bound is tight because the latent was never used.

Practice: Set, problem C

Give the inverse and the log-absolute-determinant of $y_1 = h_1$, $y_2 = h_2 e^{h_1} + h_1^2$.

Worked solution

$h_1 = y_1$ and $h_2 = (y_2 - y_1^2)e^{-y_1}$. The Jacobian is triangular with diagonal $(1, e^{h_1})$, so the log-absolute-determinant is $h_1 = y_1$.

Practice: Set, problem D

On the bin $[0, 1)$, let $p(x + u) = c\, e^u$ for $u \in [0, 1)$, with c chosen so that you do not need its value. Is uniform dequantization tight?

Worked solution

No. Tightness needs $p(x+u)$ constant in u. Here it grows with u. The optimal q is proportional to e^u on the bin.

Practice: Set, problem E

A 1×1 convolution uses $W = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ on a 3×4 feature map with 2 channels. Compute $\log|\det J|$.

Worked solution

W is a permutation, so $|\det W| = 1$ and the log-determinant is zero. The spatial factor $HW = 12$ multiplies zero. Permutations mix channels and do not change volume. Glow's learned diagonal s is what changes volume.

References and slide index

The lectures are the primary source. The books and papers below are the ones named on Lecture 0, slide 12, on Lecture 5, slide 84, and in the homework sheets.

- Sajjad Amini, Deep Generative Models lecture decks. Lecture 2, slides 12–42, for densities, expectation, Monte Carlo, and KL. Lecture 3 for autoregressive models and MADE. Lecture 4, slides 12–40, 56–64, and 77–84, for the ELBO and the reparameterization trick. Lecture 5, slides 3–73, for change of variables, NICE, RealNVP, and MAF.
- Kevin P. Murphy, *Probabilistic Machine Learning: Advanced Topics*, MIT Press, 2023. Chapter 10 for variational inference, Chapter 11 for Monte Carlo and importance sampling, Chapter 21 for variational autoencoders, Chapter 23 for normalizing flows.
- Jakub M. Tomczak, *Deep Generative Modeling*, Springer. Section 4.1.2 for the change of variables, §4.1.3 for RealNVP and dequantization, §5.3 for the VAE, §5.3.6 for posterior collapse, §5.4.1.4 for a mixture prior, §5.4.1.8 for a flow prior, §5.5.2.1 for a two-level hierarchical VAE.
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- M. Germain, K. Gregor, I. Murray, and H. Larochelle, MADE: Masked Autoencoder for Distribution Estimation, ICML 2015.

Intuition: A path through the two homeworks

Read Chapters 2–4 once. Do Homework 1's Gaussian and divergence problems from Chapters 5 and 6, and the autoregressive problems from Chapter 7. Then read Chapter 8 and write the conditional ELBO before you train the VAE notebook. Read Chapter 9 before problem 2. Read the coupling section of Chapter 10 before problems 3, 4, and 5, and before the flow notebook. Do the notebook's color-transfer section first. It is the two-dimensional MAF.